

WeatherLink Live Weather Station 2.1.0

Version 2.1.000.0000

Driver

WeatherLink Live Weather Station is a Crestron Home extension driver for current weather conditions and forecast information.

The tile appears on the Home screen and opens the current conditions page.

Notes and Recommendations

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WeatherLink and WeatherLink Live are trademarks of Davis Instruments Corp. This independent project is not affiliated with, endorsed by, or sponsored by Davis Instruments Corp.

OpenWeather provides forecast data and optional cloud fallback current conditions. This independent project is not affiliated with, endorsed by, or sponsored by OpenWeather.

System Requirements and Dependencies

- A compatible Crestron Home processor, plus either a WeatherLink Live device or cloud-only OpenWeather operation.
- WeatherLink Live Host: optional IP address or hostname for local current conditions.
- OpenWeather API Key: required in every operating mode. A free OpenWeather account is sufficient for current conditions and the reduced five-day forecast. The driver automatically uses the available supported OpenWeather service for the account key. Existing One Call API 3.0 subscriptions remain usable; new customers can use an eligible One Call API 4.0 subscription with the same account key. No separate SimpleWeatherClient key is needed.
- Location Name Override: optional title location name.
- Latitude Override and Longitude Override: optional paired coordinates for cloud weather requests.
- Units: Metric, UK, or Imperial.
- Refresh Interval Seconds: persistent current-condition interval from 60 to 86400 seconds, default 300.

Installation/Upgrade Instructions

Install the packaged driver in Crestron Home, add WeatherLink Live Weather Station, and enter its connection settings. WeatherLink Live Host is optional; when local readings are unavailable the driver can use cloud data for fallback current conditions. Leave the host blank for cloud-only operation, which uses OpenWeather for both current conditions and forecasts.

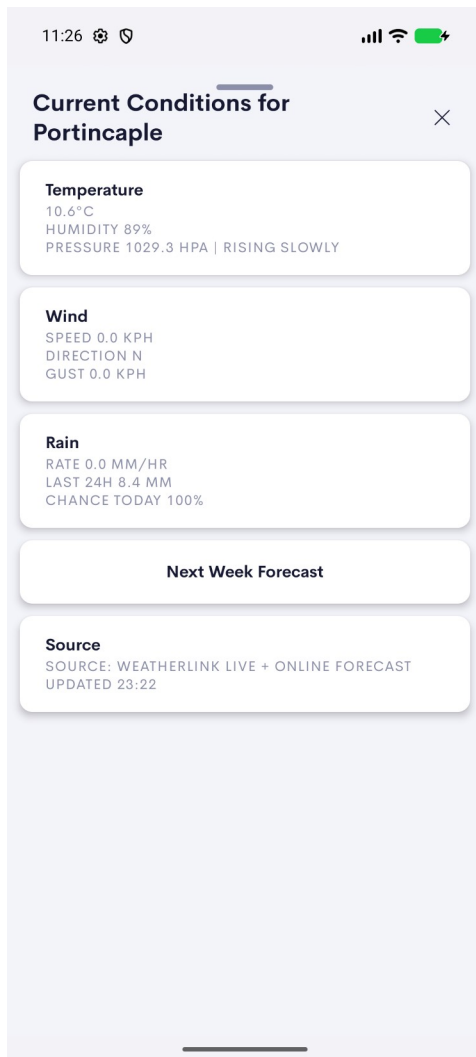
Enter the OpenWeather API Key for forecast and cloud fallback, or for cloud-only operation. Cloud-only operation requires Internet access and the processor location or paired coordinate overrides. A WeatherLink Live device alone cannot be used without an OpenWeather API Key in this release. Service selection is automatic; there is no per-controller setting for it. Set Location Name Override only when a different displayed location name is wanted.

Latitude Override and Longitude Override must be supplied together. When both are blank, the driver uses the processor location. Select Metric, UK, or Imperial units, then set Refresh Interval Seconds within the documented range.

End-User Experience

Current conditions

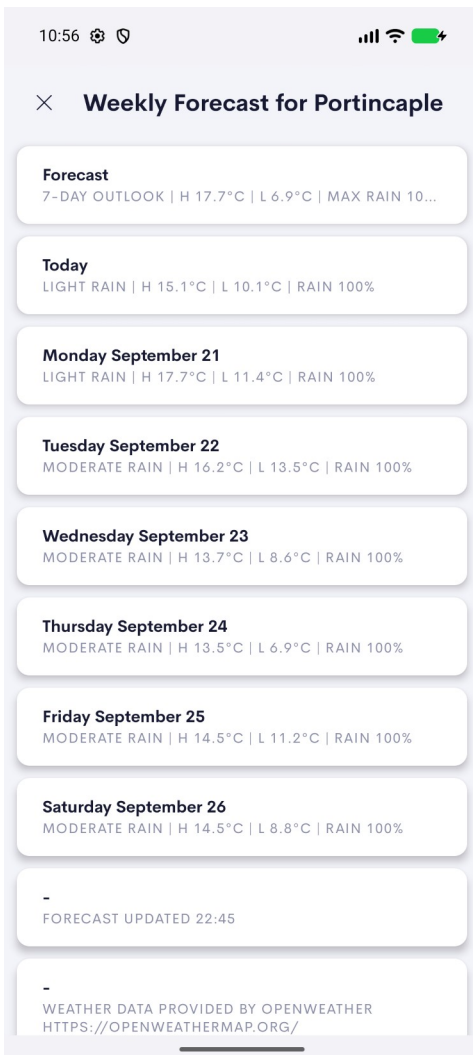
The current conditions page displays temperature, wind, rainfall, pressure, humidity and the data source. Values use the selected unit system. Select Next Week Forecast to open the forecast page. Close the page to return to Home.



Current conditions from WeatherLink Live, with humidity, pressure, wind and rainfall. Example readings are shown.

Weekly forecast

The weekly forecast page displays a summary and seven daily entries, including the weather description, high and low temperature, and chance of rain. The bottom of the page shows the forecast update time and OpenWeather attribution. Scroll when needed to see the remaining entries. Close the forecast to return to current conditions.



Weekly forecast with daily weather, temperatures and rain probability. Example readings are shown.

Limitations/Known Issues

Forecast and cloud fallback depend on network access and an OpenWeather service available to the configured account key. Automatic selection retains an existing One Call API 3.0 subscription, tries an eligible One Call API 4.0 subscription after access denial, then uses the free current-conditions and reduced five-day forecast endpoints after access denial. Timeouts, rate limits and server failures do not change service. Failed cloud attempts are temporarily throttled. The Home tile is not shown on room screens.

Online status describes available current-weather data. A working cloud fallback can keep the driver online while the local station is unreachable. If current-weather retrieval fails with no usable fallback, the driver goes offline and retains any last available readings with a failed-update status. Online status returns after a successful current-weather refresh. A forecast-only failure does not make fresh local current conditions offline.

Supported Features

- Local WeatherLink Live current conditions when available.
- Forecast information and cloud fallback current conditions.
- Metric, UK, and Imperial unit selections.

Test Environment

Observed test environment: CP4-R (hardware version 2), processor firmware 2.8000.00057, and Crestron Home Android app 4.11.1+pr on an Android 16 emulator (API level 36). Current conditions were obtained from one WeatherLink Live unit; forecast data came from OpenWeather. The weather-device firmware version was not recorded.

Build environment: .NET SDK 10.0.401 targeting .NET Framework 4.7.2, Crestron Driver SDK runtime references 27.0.24, and ManifestUtil 29.0.10. The screenshots illustrate the interface; their weather readings are examples and change over time.

Supported Models

WeatherLink Live provides local current conditions when configured. The package declares WeatherLink Live Weather Station as its model and WeatherLink Live as its supported series; no additional models are declared. A WeatherLink Live device is optional: OpenWeather supplies both current conditions and forecast data in cloud-only operation.

Contact Information

Support: <https://oznetmaster.github.io/support/>

Source and documentation: <https://github.com/oznetmaster/WeatherLinkLiveCrestronDriver>

Version History

2.0.19: Documentation-only update. Corrected the bundled help version-history and dependency/license descriptions. Runtime behavior, configuration, commands, and UI are unchanged.

2.0.18: Updates WeatherLinkLiveLibrary and SimpleWeatherClient to 2.0.0, adding automatic OpenWeather service compatibility for eligible One Call API 3.0 or 4.0 accounts and free-endpoint fallback.

2.0.17: Corrected offline reporting when current-weather sources fail, retained last readings during failures, and preserved online status for fresh local conditions when only forecast retrieval fails.

2.0.16: Corrected displayed pressure, cloud fallback wind speed and imperial rainfall units. Updated embedded help, dependency notices and package resource loading.

Licensing and Copyright Information

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The merged components include WeatherLinkLiveLibrary 2.1.0, SimpleWeatherClient 2.0.0, System.Text.Json and its Microsoft compatibility libraries, and Hafner compatibility attributes. They retain their own licenses and copyright notices; the package notice file includes the reviewed texts and identifies the component versions.

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Connection recovery and programmable events

After a previously working local station becomes unavailable, the driver retains its client and lets it reconnect automatically. Retry delays are 10, 20, 30, 40, 50, 60 and 120 seconds, then 300 seconds repeatedly. Each delay starts after the previous request fails. Success restores local readings immediately and resets the retry sequence. Initial connection failures continue to use the configured refresh behavior.

Use these events in Crestron Home programming:

WeatherLink Disconnected (weatherLinkDisconnected): fires once when a previously working local station stops returning valid readings.

WeatherLink Reconnected (weatherLinkReconnected): fires once after fresh local readings return. Initial connection does not fire this event.

These events concern the local WeatherLink station. Successful cloud fallback can keep the overall driver online while the local station is disconnected. Forecast failures do not make fresh local readings offline. Recovery is independent of the normal current-condition refresh interval.

Upgrade and validation

Driver identity, existing configuration fields, commands and UI remain unchanged. This version uses WeatherLinkLiveLibrary 2.1.0 and SimpleWeatherClient 2.0.0. It preserves current unit selections when reusing a client and prevents late cloud responses from replacing recovered local readings.

Validation included 90 processor unit/lifecycle checks, three real-station reading checks and a physical station network-disconnect/reconnect test. This is recovery validation, not a new endurance run. The separate processor test package uses NUnit 5; it is not required for normal driver operation.

Version history

2.1.0 - 1 October 2026: automatic local recovery, programmable disconnect/reconnect events and immediate restored readings. Earlier version history is on page 5.

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